Excel

Formulas | Lookups | Pivot Tables | Dashboards

Quick notes + interview questions and answers for Data Analyst roles. Short answers you can say out loud, with formulas you can write on the spot.

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Part 1 - Formula Basics

Formulas always start with =. Master references, logical and aggregate functions first; everything else builds on them.

Quick Notes

- Relative (A1) changes when copied; absolute ($A$1) stays fixed; mixed ($A1 or A$1) locks only one part. F4 cycles through them.
- Order of operations: brackets, then ^, then * and /, then + and -.
- Use Ctrl+Shift+Enter only for old array formulas; modern Excel spills dynamic arrays automatically.
- Common errors: #DIV/0! (divide by zero), #N/A (lookup not found), #REF! (deleted cell), #VALUE! (wrong data type), #NAME? (typo in function).

Q1. Relative vs absolute vs mixed references?

Answer: A relative reference (A1) shifts when you copy the formula. An absolute reference ($A$1) never changes. A mixed reference locks only the row (A$1) or only the column ($A1). Press F4 to toggle between them.

```
=B2*$E$1      'tax rate in E1 stays fixed while B2 moves down
=$A2*B$1      'multiplication-table style
```

Q2. SUM vs SUMIF vs SUMIFS?

Answer: SUM adds all numbers in a range. SUMIF adds numbers that meet one condition. SUMIFS adds numbers that meet many conditions; note that its sum range comes first.

```
=SUMIF(B:B,"Delhi",C:C)
=SUMIFS(C:C, B:B,"Delhi", D:D,">=1000")
```

Q3. COUNT vs COUNTA vs COUNTBLANK vs COUNTIFS?

Answer: COUNT counts numeric cells, COUNTA counts non-empty cells, COUNTBLANK counts empty cells, and COUNTIFS counts cells that meet multiple conditions.

```
=COUNTA(A2:A100)
=COUNTIFS(B:B,"ER", C:C,">30")
```

Q4. How does IF work, and how do you nest it?

Answer: IF(test, value_if_true, value_if_false). Nest IFs for several outcomes, or use IFS / SWITCH for cleaner formulas. Combine conditions with AND and OR.

```
=IF(C2>=90,"A",IF(C2>=75,"B","C"))
=IFS(C2>=90,"A",C2>=75,"B",TRUE,"C")
=IF(AND(B2>0,C2<30),"OK","Check")
```

Q5. How do you handle errors in formulas?

Answer: Wrap the formula in IFERROR(value, fallback) to show a friendly result. Use IFNA to catch only #N/A, so you do not hide real mistakes.

```
=IFERROR(B2/C2,0)
=IFNA(VLOOKUP(A2,Data!A:C,3,FALSE),"Not found")
```

Q6. AVERAGE vs MEDIAN vs MODE - when to use which?

Answer: Average is affected by extreme values. Median is the middle value and is better when there are outliers (salaries, wait times). Mode is the most frequent value, useful for categories or sizes.

```
=AVERAGE(C2:C100)
=MEDIAN(C2:C100)
=MODE.SNGL(C2:C100)
```

Q7. How do you find the largest, smallest or nth value?

Answer: Use MAX and MIN for extremes, LARGE(range,n) and SMALL(range,n) for the nth value, and RANK.EQ for the rank of a value.

```
=LARGE(C2:C100,3)
=RANK.EQ(C2,$C$2:$C$100,0)
```

Q8. What are named ranges and why use them?

Answer: A named range gives a cell or range a readable name (for example TaxRate), so formulas are easier to read and less error-prone. Create one from Formulas > Define Name or the Name Box.

```
=SUM(Sales_2026)
=B2*TaxRate
```

Q9. What is a dynamic array and what are FILTER, SORT, UNIQUE?

Answer: In Excel 365 a formula can return many values that spill into neighbouring cells. FILTER returns rows that meet a condition, SORT orders them and UNIQUE returns distinct values.

```
=UNIQUE(B2:B500)
=SORT(FILTER(A2:D500,B2:B500="ER"),4,-1)
```

Q10. What is the difference between a formula and a function?

Answer: A formula is any expression starting with = (it can include numbers, cell references and operators). A function is a built-in routine used inside a formula, such as SUM or VLOOKUP.

Part 2 - Lookup Functions

Lookups pull matching data from another table, like a join in SQL. Know all four and when to prefer each.

Quick Notes

- VLOOKUP looks right only. XLOOKUP looks any direction and defaults to exact match.
- INDEX-MATCH is older but works in every version and is flexible.
- Always use exact match (FALSE / 0) unless you really want an approximate match on sorted data.
- Lookup values must have the same type and no hidden spaces: use TRIM when matches fail.

Q11. How does VLOOKUP work and what are its limits?

Answer: VLOOKUP(lookup_value, table, col_index, FALSE) searches the first column of the table and returns a value from the column number you give. Limits: it looks only to the right, breaks if columns are inserted (hard-coded index), and returns only the first match.

```
=VLOOKUP(A2, Products!A:C, 3, FALSE)
```

Q12. What is XLOOKUP and why is it better?

Answer: XLOOKUP(lookup, lookup_array, return_array, if_not_found) searches any column, returns from any column (left or right), is exact match by default, has a built-in not-found message and can search from last to first.

```
=XLOOKUP(A2, Products[ID], Products[Price], "Not found")
```

Q13. How do you use INDEX-MATCH?

Answer: MATCH finds the position of a value in a range; INDEX returns the value at that position of another range. Together they replace VLOOKUP and work in any direction.

```
=INDEX(C:C, MATCH(A2, A:A, 0))
```

Q14. VLOOKUP vs INDEX-MATCH vs XLOOKUP - which would you choose?

Answer: XLOOKUP if the version supports it (simplest and safest). INDEX-MATCH when files must work in older Excel or for two-way lookups. VLOOKUP only for quick, simple lookups where the key is the first column.

Q15. Why does my VLOOKUP return #N/A even though the value exists?

Answer: Common reasons: extra spaces or different data types (text vs number), approximate match used by mistake, wrong table range that does not include the key column first, or the lookup value really is missing. Fix with TRIM, VALUE or TEXT and use FALSE for exact match.

```
=VLOOKUP(TRIM(A2), Data!A:C, 2, FALSE)
```

Q16. How do you do a two-way lookup?

Answer: Use INDEX with two MATCH functions, one for the row and one for the column.

```
=INDEX(B2:M20, MATCH(A23,A2:A20,0), MATCH(B22,B1:M1,0))
```

Part 3 - Pivot Tables and Charts

A pivot table summarises thousands of rows in seconds without writing formulas. It is the most asked Excel topic for analyst roles.

Quick Notes

- Source data must be a clean table: one header row, no blank rows or columns, no merged cells. Convert to a Table with Ctrl+T.
- Four areas: Rows, Columns, Values, Filters. Drag fields between them to reshape the report.
- After the source data changes, press Refresh (Alt+F5); pivots do not update by themselves.
- Slicers and Timelines are visual filters you can connect to several pivots.

Q17. What is a pivot table and how do you create one?

Answer: A pivot table summarises, groups and cross-tabulates a dataset interactively. Click inside the data, choose Insert > PivotTable, then drag fields to Rows, Columns, Values and Filters.

Q18. How do you group dates or numbers in a pivot?

Answer: Right-click a date field > Group, then choose months, quarters or years. For numbers, group into bins (for example age bands of 10).

Q19. What is a calculated field in a pivot table?

Answer: A formula-based field you add inside the pivot using other fields, for example Profit = Sales - Cost. It works on the summed values, so ratios can be misleading; use it with care.

Q20. Pivot table vs formulas like SUMIFS?

Answer: A pivot is fast to build and easy to explore but needs a manual refresh. SUMIFS formulas update automatically and sit exactly where you place them but take more effort to set up. For dashboards I often combine both.

Q21. What are slicers and timelines?

Answer: Slicers are clickable buttons that filter pivot tables and charts. Timelines are a date-specific slicer for filtering by month, quarter or year. Both can be linked to multiple pivots.

Q22. Which chart do you pick for which data?

Answer: Line for trends over time, column or bar to compare categories, pie or doughnut only for a few parts of a whole, scatter for relationships, combo chart for two measures with different scales, histogram or box plot for distributions.

Q23. How do you build a combo chart with a secondary axis?

Answer: Select the data, insert a chart, change the series type (Change Chart Type > Combo), set one series to a line and tick Secondary Axis. Use it when two measures have very different scales, for example revenue and growth percent.

Q24. How do you make a chart clear for stakeholders?

Answer: One message per chart, a clear title that states the insight, labelled axes with units, no unnecessary gridlines or 3-D effects, a consistent colour scheme and highlighted key values with data labels.

Part 4 - Data Cleaning, Formatting and Validation

Real data is messy. Good analysts clean first, then analyse.

Quick Notes

- Cleaning toolkit: Remove Duplicates, TRIM, CLEAN, Text to Columns, Find and Replace, Flash Fill (Ctrl+E), Go To Special (blanks).
- Convert numbers stored as text with VALUE or Paste Special > Multiply by 1.
- Never overwrite raw data: keep the original sheet and clean in a copy.

Q25. How do you remove duplicates?

Answer: Select the data > Data > Remove Duplicates and choose the columns that define a duplicate. In Excel 365 you can also use UNIQUE to get distinct rows without touching the original.

```
=UNIQUE(A2:C500)
```

Q26. Which functions clean text?

Answer: TRIM removes extra spaces, CLEAN removes non-printing characters, UPPER / LOWER / PROPER change case, LEFT / RIGHT / MID extract parts, SUBSTITUTE replaces text, TEXTJOIN / CONCAT join text.

```
=PROPER(TRIM(A2))
=LEFT(A2,FIND(" ",A2)-1)
=SUBSTITUTE(A2,"-","")
```

Q27. How do you split one column into several?

Answer: Use Data > Text to Columns (delimiter or fixed width), Flash Fill (Ctrl+E) for pattern-based splitting, or TEXTSPLIT in Excel 365.

```
=TEXTSPLIT(A2," ")
```

Q28. How do you work with dates in Excel?

Answer: Dates are serial numbers, so you can add and subtract them. Use YEAR, MONTH, DAY, EOMONTH, DATEDIF, NETWORKDAYS and TEXT for formatting.

```
=DATEDIF(A2,B2,"d")
=TEXT(A2,"mmm-yyyy")
=NETWORKDAYS(A2,B2)
```

Q29. What is conditional formatting and where do you use it?

Answer: It formats cells automatically based on rules (colour scales, data bars, icon sets or a custom formula). I use it to highlight outliers, overdue items, duplicates and top / bottom values.

```
Custom rule:  =$D2>30      'highlight rows where wait time > 30
```

Q30. What is data validation?

Answer: It restricts what can be typed into a cell: whole numbers in a range, a date range, or a drop-down list. It prevents bad data at the source and supports dependent drop-downs.

```
Data > Data Validation > List > Source:  ER,ICU,OPD
```

Q31. How do you find and fill blank cells quickly?

Answer: Select the range, press Ctrl+G > Special > Blanks, type = and the cell above (or a value) and press Ctrl+Enter to fill all selected blanks at once.

Q32. What is Power Query and when do you use it?

Answer: Power Query (Data > Get Data) imports, cleans and transforms data with repeatable steps: remove columns, split, merge, unpivot, change types. When the source updates, press Refresh and all the steps run again, so it replaces manual cleaning.

Part 5 - Dashboards and Scenario Questions

Interviewers test how you think and communicate, not only formulas.

Q33. How do you build an Excel dashboard?

Answer: 1) Clean the data and convert it to a Table. 2) Create pivot tables for each KPI or view. 3) Insert charts from the pivots. 4) Add slicers connected to all pivots. 5) Arrange KPI cards, charts and filters on one sheet. 6) Hide gridlines, apply a consistent theme and test the filters.

Q34. Walk me through your Hospital Emergency Room Dashboard project.

Answer: I used patient-visit data to track visits, wait times, admissions and department-wise performance. I cleaned the data, built pivot tables for each metric, made charts such as visits over time and average wait time by department, added slicers for date and department, and arranged KPI cards on one page so a manager can find bottlenecks quickly.

Q35. What KPIs would you show for an ER dashboard?

Answer: Total patient visits, average wait time, admission rate, visits by department, peak hours and days, and the share of patients waiting longer than a target (for example 30 minutes).

Q36. What are the most useful Excel shortcuts?

Answer: Ctrl+T (table), Ctrl+Shift+L (filter), Ctrl+Arrow (jump to edge), Ctrl+Shift+Arrow (select to edge), F4 (lock reference), Ctrl+E (Flash Fill), Alt+= (AutoSum), Ctrl+1 (format cells), Alt+F5 (refresh pivot).

Q37. Excel vs SQL vs Python - when would you use each?

Answer: Excel for quick analysis, small to medium data and visual dashboards. SQL for querying and aggregating data stored in databases. Python for large data, automation, advanced cleaning and statistical work. In practice they work together.

Q38. How do you check that your numbers are correct?

Answer: Reconcile totals with the source (row counts, sums), cross-check a few values manually, compare pivot totals with SUMIFS results, look for blanks or duplicates, and review outliers before sharing the report.

Q39. Your sales file has 5 lakh rows and Excel is slow. What do you do?

Answer: Convert formulas to values where possible, remove unused columns and formats, use a Table or Power Query to load only needed data, use pivot tables instead of many SUMIFS, and switch to manual calculation while editing. If it is still too large, move the data to a database or Python.

Last-minute Cheat Sheet

Task	Formula / Shortcut
Conditional sum	`=SUMIFS(sum_rng, crit_rng1, crit1, crit_rng2, crit2)`
Conditional count	`=COUNTIFS(rng1, crit1, rng2, crit2)`
Lookup (modern)	`=XLOOKUP(key, key_rng, return_rng, "Not found")`
Lookup (any version)	`=INDEX(return_rng, MATCH(key, key_rng, 0))`
Error handling	`=IFERROR(formula, 0)`
Multi-condition IF	`=IFS(c1,v1, c2,v2, TRUE,v3)`
Clean text	`=PROPER(TRIM(A2))`
Date difference	`=DATEDIF(start, end, "d")`
Distinct list	`=UNIQUE(range)`
Filter rows	`=FILTER(range, cond)`
Table / filter	`Ctrl+T / Ctrl+Shift+L`
Lock reference	`F4`
Refresh pivot	`Alt+F5`